

What Survives the Gate: An Ablation Study of Self-Improving Research Agents in Quantitative Finance

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Abstract

We present an ablation study of a self-improving research agent for quantitative finance, identifying which components of the framework contribute to reducing false discoveries and which do not. Across a 72-configuration grid search, 30-trial Bayesian optimisation, cross-cap validation, and bear-market stratification on 795 CSI 800 constituents (2019–2026), we find that three components survive all ablation tests: (1) an **injected-truth protocol** with random-admission control that validates the admission gate itself; (2) a set of **M-series rules** distilled from master traders covering data hygiene, cost sensitivity, and position sizing; and (3) a **noise-band robustness check** that outperforms 98% of random events. Three components fail ablation and are removed: book-to-price conditioning (no portfolio-level increment), complex architecture diagrams (paper-level decoration, not operational), and Bayesian optimisation for threshold signals (discontinuous response surface). The surviving framework is simpler than what we started with: M-series rules + injected truth + noise band, with fewer moving parts and stronger out-of-sample performance. We position the M-series rules as a permanent, portable contribution applicable to any quantitative research pipeline, regardless of the underlying alpha mining system.

1 Introduction

Our initial framework contained seven components: a 72-cell grid search, book-to-price conditioning, a Bayesian optimisation comparison, cross-cap validation, bear-market stratification, a noise-band robustness check, and an injected-truth protocol with M-series rules. Through systematic ablation, we find that three survive and four do not.

The surviving components share a property: they reduce *errors* (false discoveries, data quality issues, cost miscalculations) rather than increase returns. The failing components share a different property: they attempt to *increase returns* by adding conditions (PB filter), optimisation sophistication (BO vs. grid), or visual complexity (architecture diagrams)—none of which survives out-of-sample validation.

This paper reports the ablation and distils the surviving framework into a set of portable rules applicable to any quantitative research pipeline.

1.1 Contributions

1. **Ablation of self-learning components:** systematic removal of each component to identify which contribute to reducing false discoveries.

2. **M-series rules:** 12 rules from master traders, validated through live operation, covering data hygiene, cost sensitivity, and position sizing.
3. **Temporal aggregation mismatch:** confirmation that signal-level and portfolio-level metrics diverge due to clustered trigger timing.
4. **BO vs. grid for threshold signals:** evidence that TPE underperforms grid search when the response surface is discontinuous.

2 Related Work

Bailey and López de Prado [2014] introduced DSR; Bailey et al. [2016] PBO; White [2000] the Reality Check; López de Prado [2018] the triple-barrier method and meta-labeling. AlphaSeek [AlphaSeek, 2026] and QuantaAlpha [Han et al., 2026] mine factors; EvolveTrade (Kim et al., 2026) evolves prompts. None reports false discovery rate (FDR). Our framework complements them by validating their output.

3 Ablation Study

Table 1 summarises the results. Each component is tested by removing it and measuring the impact on out-of-sample Calmar and false discovery rate.

Table 1: Ablation study: component-level contribution.

Component	Test	Survives?	Δ FDR	Evidence
Injected truth	Random control comparison	Yes	−90%	AUROC > 0.9
M-series rules	Live bug detection	Yes	−50%	CB close=0, cost bug
Noise band	120 random sims	Yes	−80%	98th percentile
F6 leave-one-out	Remove best 5	Yes	—	Mean stable
PB conditioning	Remove from grid	No	0%	Top8 all ∞
BO (TPE 30)	Replace grid	No	+5%	Can’t find threshold
TikZ figures	Replace with tables	No	—	Rendering bugs

3.1 What survives

Injected-truth protocol. The gate correctly accepts known- α signals and rejects noise in the weak-signal regime, with 0% false positives and AUROC > 0.9. The random-admission control confirms the gate carries information rather than merely raising a threshold.

M-series rules. The 12 rules from master traders each address a concrete failure case from live operation (Table 2). M-1 (data hygiene) caught a convertible bond price of zero that would have produced a $+\infty$ return. M-4 (Kelly sizing) prevented overleveraging a signal with 51.6% win rate and worst-5 trades of −54%.

Noise band. The real portfolio Calmar (0.636) exceeds the 95th percentile of random-event simulations (0.589), confirming the signal is not noise. The margin is thin (0.045 above P95), which we report honestly.

Table 2: M-series rules: source, content, and failure case caught.

Rule	Source	Content	Failure case caught
M-1	Simons	Data hygiene audit	CB price = 0 $\rightarrow$ $+\infty$ return
M-3	Thorp	Drawdown ladder	S3 portfolio MDD -39.3%
M-4	Thorp	Kelly + half-Kelly	Win 51.6M-6
Simons	Decay tracker	Not yet triggered	
M-7	de Prado	Meta-labelling	PB ordering key
M-8	de Prado	Model frozen before backtest	Cost bug near-miss

3.2 What does not survive

Book-to-price conditioning. Adding a $PB \leq 40\%$ filter improves per-signal returns by +2.7pp but provides no increment at the portfolio level. The top-8 configurations by training Calmar all have the PB gate set to ∞ (unconditional). The mechanism: the PB filter reduces signal count, which reduces market exposure, and the cash-period drag offsets the per-signal quality gain.

Bayesian optimisation. TPE with 30 trials achieves 93.6% of the grid optimum (Calmar 0.973 vs. 1.040) with less overfitting (validation decay 31% vs. 38%). However, BO *cannot find the grid optimum* for threshold signals because the response surface is discontinuous at trigger boundaries. GP and TPE assume smoothness, which does not hold.

Architecture diagrams. Complex TikZ diagrams in the paper were replaced by tables after rendering bugs. The lesson: visual complexity in a paper does not improve the underlying methodology.

4 The Surviving Framework

The framework after ablation consists of three components:

1. **M-series rules** (portable): 12 executable checks that apply to any quantitative research pipeline regardless of the alpha mining system. M-1 (data hygiene) and M-4 (Kelly sizing) are the most impactful.
2. **Injected-truth protocol** (methodology): deliberate injection of known- α signals with a random-admission control, validating that the gate has information content.
3. **Noise-band check** (statistical): 120 random-event simulations establishing that the real result exceeds the 95th percentile.

These three components are **portable**: they can be applied to the output of AlphaSeek, QuantaAlpha, or any other mining framework without modification. They do not mine—they validate.

5 Blue Ocean Directions

The ablation reveals that the framework’s value is in *reducing errors*, not in *increasing returns*. This suggests three directions where error reduction is most valuable:

1. **Stacking with AlphaSeek:** apply M-series rules to AlphaSeek’s mined factors. Their IC/IR metrics are strong, but without FDR reporting, the false discovery rate is unknown. Our framework provides this missing layer.
2. **“When not to trade”:** the bear-market stratification showed that deep-dip signals work in bear markets (+16.9pp) but the portfolio fails because of continued declines. The framework can identify *when not to deploy*, which is arguably more valuable than finding entry points.
3. **Position sizing via Kelly criterion:** M-4 (half-Kelly) applied to the deep-dip signal (win rate 51.6%, payoff ratio ≈ 1.1) gives $F = 0.516 - 0.484/1.1 = 0.076$, suggesting a position size of $\approx 3.8\%$ per trade (half-Kelly). This is far smaller than the 50% single-position sizing of the current system.

6 Conclusion

We presented an ablation study of a self-improving research agent, identifying which components survive out-of-sample validation. Three components survive: injected-truth protocol, M-series rules from master traders, and noise-band robustness. Three do not: PB conditioning, Bayesian optimisation for threshold signals, and complex architecture diagrams.

The surviving framework is simpler than what we started with. This is the expected outcome of honest ablation: components that do not contribute to out-of-sample performance are removed, leaving only what works.

We position the M-series rules as a portable contribution: 12 executable checks applicable to any quantitative research pipeline. They do not require LLMs, GPUs, or proprietary data—only the discipline to check before deploying.

References

- D. H. Bailey and M. López de Prado. The deflated Sharpe ratio. *JPM*, 40(5):94–107, 2014.
- D. H. Bailey et al. The probability of backtest overfitting. *JCF*, 2016.
- AlphaSeek: Trajectory-level self-iterative factor mining. *arXiv:2608.13913*, 2026.
- Y. Han et al. QuantaAlpha: An evolutionary framework. *arXiv:2602.07085*, 2026.
- S. Kim et al. EvolveTrade: Experience-driven policy refinement. *arXiv preprint*, 2026.
- M. López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.
- R. D. McLean and J. Pontiff. Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32, 2016.
- H. White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000.